The Ellenberg–Gijswijt Cap-Set Upper Bound via Slice Rank

1 Preliminaries

We work over the field $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ and the vector space $\mathbb{F}_3^n$ of dimension n. Elements of $\mathbb{F}_3^n$ are written $x = (x_1, \ldots, x_n)$ with $x_i \in \{0, 1, 2\}$; addition is coordinatewise modulo 3.

Throughout, $C, c, C_1, C_2, \ldots$ denote positive absolute constants, independent of n, whose values may change from line to line.

1.1 Cap sets

Definition 1.1 (Cap set)**.** A subset $A \subseteq \mathbb{F}_3^n$ is called a *cap set* (or 3-*AP-free*) if it contains no three-term arithmetic progression with distinct terms; equivalently, the only solutions $(x, y, z) \in A \times A \times A$ to $x + y + z = 0$ are the trivial ones $x = y = z$.

Remark 1.2. The equivalence in Definition 1.1 is specific to $\mathbb{F}_3$: in characteristic 3, the relation $x+y+z = 0$ rewrites as $z = -x-y = 2x+2y$, and three-term arithmetic progressions $(x, x+d, x+2d)$ satisfy $x + (x + d) + (x + 2d) = 3x + 3d = 0$ exactly. A degenerate solution $x = y = z$ corresponds to the constant progression $d = 0$.

1.2 The monomial space and the count M_n

The reduced polynomial ring is

$$R_n \triangleq \mathbb{F}_3[x_1, \ldots, x_n]/\langle x_1^3 - x_1, \ldots, x_n^3 - x_n\rangle.$$

Since $x^3 = x$ in $\mathbb{F}_3$ pointwise for $x \in \mathbb{F}_3$, every function $f : \mathbb{F}_3^n \to \mathbb{F}_3$ is represented by a unique element of R_n, namely its expansion in the basis of squarefree-in-exponent monomials. A monomial in R_n takes the form $\prod_{i=1}^n x_i^{a_i}$ with $a_i \in \{0, 1, 2\}$.

Definition 1.3 (Low-degree monomial space)**.** For each $n \in \mathbb{N}$, define

$$M_n \triangleq \left|\left\{(a_1, \ldots, a_n) \in \{0, 1, 2\}^n \,:\, \textstyle\sum_{i=1}^n a_i \le 2n/3\right\}\right|,$$

the number of monomials $\prod_i x_i^{a_i}$ in R_n of total degree at most $2n/3$. We write

$$\mathcal{M}_n^{\le 2n/3} \triangleq \operatorname{span}_{\mathbb{F}_3}\left\{\prod_{i=1}^n x_i^{a_i} \,:\, a_i \in \{0, 1, 2\},\ \sum_i a_i \le 2n/3\right\} \subseteq R_n.$$

Thus $\dim_{\mathbb{F}_3} \mathcal{M}_n^{\le 2n/3} = M_n$.

Fact 1.4 (Asymptotic count)**.** $M_n^{1/n} \to 3\gamma$ as $n \to \infty$, where γ is the unique root in $(0,1)$ of an explicit entropy-optimization problem; numerically, $3\gamma < 2.7558$. Consequently, there exists an absolute constant $C > 0$ with

$$M_n \;\le\; C \cdot (2.7558)^n \qquad \text{for all } n \ge 1.$$

The proof is an elementary large-deviation / generating-function calculation; see [EG17] for the derivation.

Remark 1.5. The optimization underlying Fact 1.4 reads $3\gamma = \min_{0<\alpha<1}(1+\alpha+\alpha^2)/\alpha^{2/3}$; the minimum is attained at α^* solving $2\alpha^2+\alpha-1=0$ to lowest order, giving the cited numerical value. We do not reproduce the calculation; it is independent of the polynomial-method argument below and is treated as a black-box arithmetic fact.

1.3 Slice rank

The slice rank, introduced by Tao [Tao16] as a symmetric refinement of the Croot–Lev–Pach [CLP17] rank argument, is the key complexity measure.

Definition 1.6 (Slice)**.** Let V_1, V_2, V_3 be finite sets and let F be a field. A function $T : V_1 \times V_2 \times V_3 \to F$ is called a *slice* if at least one of the following holds:

- there exist $f : V_1 \to F$ and $g : V_2 \times V_3 \to F$ with $T(x_1,x_2,x_3) = f(x_1)\, g(x_2,x_3)$ (a 1-*slice*);
- there exist $f : V_2 \to F$ and $g : V_1 \times V_3 \to F$ with $T(x_1,x_2,x_3) = f(x_2)\, g(x_1,x_3)$ (a 2-*slice*);
- there exist $f : V_3 \to F$ and $g : V_1 \times V_2 \to F$ with $T(x_1,x_2,x_3) = f(x_3)\, g(x_1,x_2)$ (a 3-*slice*).

Definition 1.7 (Slice rank)**.** For $T : V_1 \times V_2 \times V_3 \to F$, the *slice rank* $\operatorname{slice\,rank}(T)$ is the smallest non-negative integer r such that $T = \sum_{j=1}^{r} T_j$ with each T_j a slice (Definition 1.6); each summand may be a slice in any of the three orientations. Equivalently, $\operatorname{slice\,rank}(T) = 0$ iff $T \equiv 0$.

Remark 1.8. Slice rank is bounded above by tensor rank (a rank-1 tensor $f(x_1)g(x_2)h(x_3)$ is a 1-slice with bivariate $g(x_2)h(x_3)$), but the two are generally distinct. Only slice rank admits the diagonal lower bound of Lemma 2.1 below.

1.4 The tensor of interest

Definition 1.9 (The cap-set tensor)**.** Let $A \subseteq \mathbb{F}_3^n$. Define $T_A : A \times A \times A \to \mathbb{F}_3$ by

$$T_A(x,y,z) \;\triangleq\; \mathbf{1}[x+y+z=0],$$

where the indicator is interpreted as an element of $\mathbb{F}_3$ (so $T_A(x,y,z) = 1$ if $x+y+z=0$ in $\mathbb{F}_3^n$, and $T_A(x,y,z) = 0$ otherwise).

2 Slice rank lower bound for diagonal tensors

The following lemma is the key linear-algebraic input. It says that a tensor supported exactly on the diagonal of V^3 with nonzero diagonal entries has slice rank equal to $|V|$. We follow Tao's symmetric formulation [Tao16].

Lemma 2.1 (Diagonal lower bound for slice rank)**.** *Let V be a finite set, let F be a field, and let $T : V \times V \times V \to F$ satisfy*

(i) $T(x,y,z)=0$ *whenever* (x,y,z) *is not on the diagonal, i.e. whenever* x,y,z *are not all equal;*

(ii) $T(x,x,x)\neq 0$ *for every* $x\in V$.

Then $\operatorname{slice\,rank}(T)=|V|$.

Proof. The upper bound $\operatorname{slice\,rank}(T)\le |V|$ is immediate from the diagonal decomposition: for each $x_0\in V$ define $f_{x_0}:V\to F$ by $f_{x_0}(x)=\mathbf{1}[x=x_0]$ and $g_{x_0}:V\times V\to F$ by $g_{x_0}(y,z)=T(x_0,y,z)$. Then $T(x,y,z)=\sum_{x_0\in V} f_{x_0}(x)\,g_{x_0}(y,z)$, a sum of $|V|$ many 1-slices.

The lower bound $\operatorname{slice\,rank}(T)\ge |V|$ is the content of the lemma. The case $|V|=0$ is trivial (both sides are 0); we assume $|V|\ge 1$ henceforth. Suppose for contradiction that

$$T \;=\; \sum_{j=1}^{r_1} f_j^{(1)}(x)\,g_j^{(1)}(y,z) \;+\; \sum_{j=1}^{r_2} f_j^{(2)}(y)\,g_j^{(2)}(x,z) \;+\; \sum_{j=1}^{r_3} f_j^{(3)}(z)\,g_j^{(3)}(x,y), \tag{1}$$

with $r_1+r_2+r_3=r<|V|$. We will derive a contradiction with hypothesis Item 2.1(ii).

Eliminating the 1-slices. Let $W_1\subseteq F^V$ be the linear span of the functions $f_1^{(1)},\dots,f_{r_1}^{(1)}:V\to F$; then $\dim W_1\le r_1$. Since W_1 is a subspace of F^V of dimension at most r_1, its annihilator $W_1^{\perp}\subseteq F^V$ (under the standard bilinear form $\langle u,v\rangle=\sum_{x\in V}u(x)v(x)$) has dimension at least $|V|-r_1$.

Let $\phi:V\to F$ be any element of $W_1^{\perp}\backslash\{0\}$; equivalently, ϕ is a function on V with $\sum_{x\in V}\phi(x)f_j^{(1)}(x)=0$ for every $j\in[r_1]$. We multiply Eq. (1) by $\phi(x)$ and sum over $x\in V$. The first block vanishes by construction:

$$\sum_{x\in V}\phi(x)\sum_{j=1}^{r_1} f_j^{(1)}(x)g_j^{(1)}(y,z) \;=\; \sum_{j=1}^{r_1}\Big(\sum_{x\in V}\phi(x)f_j^{(1)}(x)\Big)g_j^{(1)}(y,z) \;=\; 0.$$

Setting $\widetilde{T}(y,z)\triangleq\sum_{x\in V}\phi(x)\,T(x,y,z)$, we therefore obtain

$$\begin{aligned}\widetilde{T}(y,z) \;&=\; \sum_{j=1}^{r_2}\Big(\sum_{x\in V}\phi(x)g_j^{(2)}(x,z)\Big)f_j^{(2)}(y) \;+\; \sum_{j=1}^{r_3}\Big(\sum_{x\in V}\phi(x)g_j^{(3)}(x,y)\Big)f_j^{(3)}(z)\\ &=\; \sum_{j=1}^{r_2}\widetilde{g}_j^{(2)}(z)\,f_j^{(2)}(y) \;+\; \sum_{j=1}^{r_3}\widetilde{g}_j^{(3)}(y)\,f_j^{(3)}(z),\end{aligned} \tag{2}$$

where $\widetilde{g}_j^{(2)}(z)\triangleq\sum_x\phi(x)g_j^{(2)}(x,z)$ and $\widetilde{g}_j^{(3)}(y)\triangleq\sum_x\phi(x)g_j^{(3)}(x,y)$. The right-hand side of Eq. (2) expresses $\widetilde{T}$ as a sum of r_2+r_3 matrix-rank-1 terms (each a function of y times a function of z, in some orientation). Hence

$$\operatorname{rank}(\widetilde{T}) \;\le\; r_2+r_3, \tag{3}$$

where $\widetilde{T}$ is viewed as a $V\times V$ matrix indexed by (y,z).

The diagonal structure of $\widetilde{T}$. By hypothesis Item 2.1(i), $T(x,y,z)=0$ unless $x=y=z$, so

$$\widetilde{T}(y,z) \;=\; \sum_{x\in V}\phi(x)T(x,y,z) \;=\; \phi(y)\,T(y,y,y)\cdot\mathbf{1}[y=z].$$

Thus $\widetilde{T}$ is a $V\times V$ diagonal matrix with diagonal entries $\phi(y)\cdot T(y,y,y)$.

Computing $\operatorname{rank}(\widetilde{T})$**.** The matrix $\widetilde{T}$ is diagonal, so its rank equals the number of nonzero diagonal entries:

$$\operatorname{rank}(\widetilde{T}) \;=\; \big|\{y \in V : \phi(y)\,T(y,y,y) \neq 0\}\big| \;=\; \big|\{y \in V : \phi(y) \neq 0\}\big|,$$

where the second equality uses hypothesis Item 2.1(ii), namely $T(y,y,y) \neq 0$ for every $y \in V$. Combining with Eq. (3),

$$|\operatorname{supp}(\phi)| \;\leq\; r_2 + r_3. \tag{4}$$

Choosing ϕ to maximize support. The inequality Eq. (4) holds for *every* $\phi \in W_1^{\perp} \setminus \{0\}$. We claim that some such ϕ has $|\operatorname{supp}(\phi)| \geq |V| - r_1$.

To see this, set $W \triangleq W_1^{\perp}$ and $d \triangleq \dim W \geq |V| - r_1$, and choose a basis $\phi_1, \ldots, \phi_d$ of W in reduced row-echelon form with respect to the standard coordinate basis of F^V. The d pivot columns lie in d distinct coordinate positions $x_1^*, \ldots, x_d^* \in V$, and by the REF property, $\phi_k(x_j^*) = \mathbf{1}[j = k]$ for $j, k \in [d]$. Set $\phi \triangleq \phi_1 + \phi_2 + \cdots + \phi_d \in W$. Then $\phi(x_k^*) = \sum_j \phi_j(x_k^*) = \phi_k(x_k^*) = 1 \neq 0$ for every $k \in [d]$. Hence $\operatorname{supp}(\phi) \supseteq \{x_1^*, \ldots, x_d^*\}$, giving $|\operatorname{supp}(\phi)| \geq d \geq |V| - r_1$.

Combining. From $|\operatorname{supp}(\phi)| \geq |V| - r_1$ and Eq. (4),

$$|V| - r_1 \;\leq\; |\operatorname{supp}(\phi)| \;\leq\; r_2 + r_3.$$

Rearranging, $|V| \leq r_1 + r_2 + r_3 = r$, contradicting the assumption $r < |V|$. This completes the proof. □

Remark 2.2. The proof works over any field, and indeed generalizes to k-tensors for any $k \geq 2$ by induction on k (with $k = 2$ being the standard rank lower bound for diagonal matrices). We use only the $k = 3$ case.

3 The no-3-AP condition forces the diagonal

We now connect the cap-set hypothesis to the diagonal structure required by Lemma 2.1.

Lemma 3.1 (Cap-set forces diagonal support)**.** *Let $A \subseteq \mathbb{F}_3^n$ be a cap set (Definition 1.1), and let $T_A : A \times A \times A \to \mathbb{F}_3$ be as in Definition 1.9. Then T_A satisfies the hypotheses of Lemma 2.1 with $V = A$. In particular,*

$$\operatorname{slice\,rank}(T_A) \;=\; |A|.$$

Proof. We verify the two hypotheses of Lemma 2.1 in turn, then invoke that lemma.

Hypothesis 2.1(ii) (nonzero diagonal). For any $x \in A \subseteq \mathbb{F}_3^n$,

$$T_A(x,x,x) \;=\; \mathbf{1}[x + x + x = 0] \;=\; \mathbf{1}[3x = 0] \;=\; \mathbf{1}[0 = 0] \;=\; 1 \;\neq\; 0 \quad \text{in } \mathbb{F}_3,$$

where the first step is the definition of T_A (Definition 1.9), the second step is $x + x + x = 3x$ as elements of $\mathbb{F}_3^n$, the third step is the identity $3x = 0$ in $\mathbb{F}_3$ (since $3 \equiv 0 \pmod 3$), and the last step is the value of the indicator at a true predicate. Thus $T_A(x,x,x) = 1$ for every $x \in A$.

Hypothesis 2.1(i) (off-diagonal vanishing). Let $(x,y,z) \in A^3$ be a triple with $T_A(x,y,z) \neq 0$. By the definition of T_A in Definition 1.9, this means $x + y + z = 0$ in $\mathbb{F}_3^n$. Since A is a cap set (Definition 1.1), the only triples $(x,y,z) \in A^3$ with $x + y + z = 0$ are the trivial ones with $x = y = z$. Therefore $T_A(x,y,z) \neq 0$ implies $x = y = z$; contrapositively, if (x,y,z) is off the diagonal, $T_A(x,y,z) = 0$. This is hypothesis Item 2.1(i).

Conclusion. Both hypotheses of Lemma 2.1 hold with $V = A$ and $T = T_A$, so $\operatorname{slice\,rank}(T_A) = |A|$, as desired. □

4 Slice rank upper bound: polynomial decomposition

The upper bound is the Croot–Lev–Pach polynomial step [CLP17], transposed to $\mathbb{F}_3^n$ and re-expressed in the symmetric slice-rank form due to Tao [Tao16]. The starting point is the polynomial identity

$$\mathbf{1}[w = 0] \;=\; 1 - w^2 \qquad \text{for } w \in \mathbb{F}_3, \tag{5}$$

which holds because $w^2 \in \{0, 1\}$ for $w \in \mathbb{F}_3$, with $w^2 = 0$ iff $w = 0$.

Lemma 4.1 (Polynomial form of T_A)**.** *For every subset $A \subseteq \mathbb{F}_3^n$ and every $(x, y, z) \in A^3$,*

$$T_A(x, y, z) \;=\; \prod_{i=1}^{n} \big(1 - (x_i + y_i + z_i)^2\big),$$

where the right-hand side is computed in $\mathbb{F}_3$.

Proof. By Definition 1.9, $T_A(x, y, z) = \mathbf{1}[x + y + z = 0]$. Since $x + y + z = 0$ in $\mathbb{F}_3^n$ iff $x_i + y_i + z_i = 0$ in $\mathbb{F}_3$ for every $i \in [n]$, and the indicator of a conjunction factors over its conjuncts,

$$\mathbf{1}[x + y + z = 0] \;=\; \prod_{i=1}^{n} \mathbf{1}[x_i + y_i + z_i = 0].$$

Applying Eq. (5) with $w = x_i + y_i + z_i$ to each factor gives the claimed identity. □

We now expand the product and split the resulting polynomial into pieces that are each slices in one of the three orientations.

Lemma 4.2 (Slice rank upper bound)**.** *For every subset $A \subseteq \mathbb{F}_3^n$,*

$$\operatorname{slice\,rank}(T_A) \;\le\; 3M_n,$$

where M_n is as in Definition 1.3.

Proof. The proof has three steps. We first expand the polynomial of Lemma 4.1 into a sum of monomials in the $3n$ variables x_i, y_i, z_i; second, we partition the monomials by which of the three variable groups $\{x_i\}$, $\{y_i\}$, $\{z_i\}$ carries the lowest total degree; third, we re-group each block into slices of one orientation each and count.

Step 1: monomial expansion. Expanding the product in Lemma 4.1 and using $(x_i + y_i + z_i)^2 = x_i^2 + y_i^2 + z_i^2 + 2x_iy_i + 2x_iz_i + 2y_iz_i$, we obtain

$$T_A(x, y, z) \;=\; \prod_{i=1}^{n} \big(1 - (x_i + y_i + z_i)^2\big) \;=\; \sum_{\alpha,\beta,\gamma} c_{\alpha,\beta,\gamma}\, x^\alpha y^\beta z^\gamma,$$

where the sum runs over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n)$, $\beta = (\beta_1, \ldots, \beta_n)$, $\gamma = (\gamma_1, \ldots, \gamma_n) \in \{0, 1, 2\}^n$, $x^\alpha = \prod_i x_i^{\alpha_i}$ (similarly y^β, z^γ), and $c_{\alpha,\beta,\gamma} \in \mathbb{F}_3$ are scalar coefficients arising from the expansion. The key structural property is:

$$c_{\alpha,\beta,\gamma} \neq 0 \;\implies\; |\alpha| + |\beta| + |\gamma| \;\le\; 2n, \tag{6}$$

where $|\alpha| = \sum_i \alpha_i$ etc. Indeed, the i-th factor $1 - (x_i + y_i + z_i)^2$, when fully expanded, contributes to each monomial a degree in $\{x_i, y_i, z_i\}$ of either 0 (the 1 term) or 2 (every monomial inside

$(x_i+y_i+z_i)^2$ has total degree 2 in the three variables x_i, y_i, z_i combined). Hence over the n factors, the combined degree in all $3n$ variables is at most $2n$ — and exactly $2k$ if k factors contributed their degree-2 part. Eq. (6) follows.

Step 2: partition by the lowest-degree variable group. For each nonzero coefficient $c_{\alpha,\beta,\gamma}$, by Eq. (6) we have $|\alpha|+|\beta|+|\gamma| \le 2n$, so at least one of $|\alpha|, |\beta|, |\gamma|$ is at most $2n/3$. Choose one such variable group; ties are broken arbitrarily but consistently (e.g., favor x over y over z). Partition the multi-index triples (α, β, γ) into three sets:

$$\begin{aligned} S_1 &\triangleq \{(\alpha,\beta,\gamma) : c_{\alpha,\beta,\gamma} \neq 0,\ |\alpha| \le 2n/3\}, \\ S_2 &\triangleq \{(\alpha,\beta,\gamma) : c_{\alpha,\beta,\gamma} \neq 0,\ |\alpha| > 2n/3,\ |\beta| \le 2n/3\}, \\ S_3 &\triangleq \{(\alpha,\beta,\gamma) : c_{\alpha,\beta,\gamma} \neq 0,\ |\alpha| > 2n/3,\ |\beta| > 2n/3,\ |\gamma| \le 2n/3\}. \end{aligned}$$

By the previous observation (at least one of $|\alpha|, |\beta|, |\gamma|$ is at most $2n/3$), every nonzero-coefficient triple lies in $S_1 \cup S_2 \cup S_3$. We accordingly split

$$T_A(x,y,z) \;=\; T^{(1)}(x,y,z) \;+\; T^{(2)}(x,y,z) \;+\; T^{(3)}(x,y,z), \tag{7}$$

where $T^{(k)}$ is the partial sum of $c_{\alpha,\beta,\gamma} x^\alpha y^\beta z^\gamma$ over $(\alpha,\beta,\gamma) \in S_k$.

Step 3: each $T^{(k)}$ is a sum of M_n slices. Consider $T^{(1)}$:

$$T^{(1)}(x,y,z) \;=\; \sum_{(\alpha,\beta,\gamma)\in S_1} c_{\alpha,\beta,\gamma}\, x^\alpha y^\beta z^\gamma \;=\; \sum_{\alpha:|\alpha|\le 2n/3} x^\alpha \cdot \Big(\sum_{(\beta,\gamma):(\alpha,\beta,\gamma)\in S_1} c_{\alpha,\beta,\gamma}\, y^\beta z^\gamma \Big),$$

where the outer sum runs over multi-indices $\alpha \in \{0,1,2\}^n$ with $|\alpha| \le 2n/3$, and the inner parenthesized expression depends only on (y,z). Each term in the outer sum is a 1-slice in the sense of Definition 1.6: $f_\alpha(x) \triangleq x^\alpha$ as a function $A \to \mathbb{F}_3$, multiplied by $g_\alpha(y,z) \triangleq \sum_{\beta,\gamma} c_{\alpha,\beta,\gamma} y^\beta z^\gamma$ as a function $A \times A \to \mathbb{F}_3$. The number of multi-indices $\alpha \in \{0,1,2\}^n$ with $|\alpha| \le 2n/3$ is exactly M_n (Definition 1.3). Hence

$$\operatorname{slice\,rank}(T^{(1)}) \;\le\; M_n. \tag{8}$$

By the same argument with the roles of x, y, z permuted: $T^{(2)}$ is a sum of 2-slices indexed by β with $|\beta| \le 2n/3$, and $T^{(3)}$ is a sum of 3-slices indexed by γ with $|\gamma| \le 2n/3$:

$$\operatorname{slice\,rank}(T^{(2)}) \;\le\; M_n, \qquad \operatorname{slice\,rank}(T^{(3)}) \;\le\; M_n. \tag{9}$$

Combining. Slice rank is subadditive under tensor addition: if $T = T' + T''$ with $\operatorname{slice\,rank}(T') \le r'$ and $\operatorname{slice\,rank}(T'') \le r''$, then T is the sum of $r' + r''$ slices and $\operatorname{slice\,rank}(T) \le r' + r''$. Applying subadditivity to Eq. (7) with the bounds in Eq. (8) and Eq. (9),

$$\operatorname{slice\,rank}(T_A) \;\le\; \operatorname{slice\,rank}(T^{(1)}) + \operatorname{slice\,rank}(T^{(2)}) + \operatorname{slice\,rank}(T^{(3)}) \;\le\; 3M_n,$$

as desired. □

Remark 4.3. The threshold $2n/3$ in the partition is forced: if the cap is at d, then for the partition to cover all monomials of total degree $\le 2n$, we need $3d \ge 2n$, i.e. $d \ge 2n/3$. The choice $d = 2n/3$ is tight; pushing it lower would invalidate the covering, pushing it higher would weaken the count M_n. This is the heart of why the $2n/3$ threshold (and the resulting base $3\gamma < 2.7558$) arises.

5 The main theorem

Theorem 5.1 (Ellenberg–Gijswijt cap-set bound)**.** *Let $n \geq 1$ and let $A \subseteq \mathbb{F}_3^n$ be a cap set (Definition 1.1). Then*

$$|A| \;\leq\; 3M_n, \tag{10}$$

where M_n is the number of monomials defined in Definition 1.3. Consequently, by Fact 1.4, there exists an absolute constant $C > 0$ such that

$$|A| \;\leq\; C \cdot (2.7558)^n \qquad \textit{for every cap set } A \subseteq \mathbb{F}_3^n. \tag{11}$$

Proof. The proof is a two-line composition of the slice-rank bounds proved above.

By Lemma 3.1 (lower bound), since A is a cap set, the tensor T_A of Definition 1.9 has slice rank exactly $|A|$:

$$\operatorname{slice\,rank}(T_A) \;=\; |A|. \tag{12}$$

By Lemma 4.2 (upper bound), regardless of whether A is a cap set,

$$\operatorname{slice\,rank}(T_A) \;\leq\; 3M_n. \tag{13}$$

Combining Eq. (12) and Eq. (13),

$$|A| \;=\; \operatorname{slice\,rank}(T_A) \;\leq\; 3M_n,$$

which is Eq. (10). The asymptotic Eq. (11) follows by substituting the bound $M_n \leq C'(2.7558)^n$ from Fact 1.4 and setting $C \triangleq 3C'$. This completes the proof. □

Remark 5.2. The exponential rate $3\gamma < 2.7558$ in Eq. (11) is the headline numerical content of [EG17]. Both [CLP17] and [EG17] match in spirit but differ in the modulus: CLP treats $\mathbb{Z}_4^n$ (cyclic group of order 4), while EG treats $\mathbb{F}_q^n$ for q odd prime power; the case $q = 3$ here is the original cap-set problem. The lower bound on cap sets (a long line of constructions starting from Edel) leaves a substantial gap below $(2.7558)^n$; closing the gap remains open as of the cited works.

6 Auxiliary cited results

For completeness we record the two external sources that frame the result of Theorem 5.1. Both are folded into the project bibliography; we do not reproduce their proofs.

Theorem 6.1 (Croot–Lev–Pach for $\mathbb{Z}_4^n$ [CLP17])**.** *There exists an absolute constant $c < 4$ such that every subset $A \subseteq (\mathbb{Z}/4\mathbb{Z})^n$ containing no three-term arithmetic progression with nonzero common difference satisfies $|A| \leq C \cdot c^n$ for some absolute constant $C > 0$.*

Theorem 6.2 (Ellenberg–Gijswijt for $\mathbb{F}_q^n$, q odd prime power [EG17])**.** *Let q be an odd prime power. Then every subset $A \subseteq \mathbb{F}_q^n$ containing no three-term arithmetic progression with distinct terms satisfies $|A| \leq 3N_q(n)$, where $N_q(n)$ is the number of monomials $\prod_{i=1}^n x_i^{a_i}$ with $0 \leq a_i \leq q-1$ and $\sum_i a_i \leq n(q-1)/3$. In particular, $N_q(n)^{1/n}$ has an explicit limit, strictly less than q, given by an entropy optimization in q variables.*

Remark 6.3. Theorem 5.1 is Theorem 6.2 specialized to $q = 3$ and reformulated via the slice-rank language of [Tao16]. The numerical bound $(2.7558)^n$ in Eq. (11) arises from 3γ at $q = 3$, where γ is the value of the entropy optimization over a discrete distribution on $\{0, 1, 2\}$.

References

[CLP17] Ernie Croot, Vsevolod F. Lev, and Péter Pál Pach. Progression-free sets in $\mathbb{Z}_4^n$ are exponentially small. *Annals of Mathematics*, 185(1):331–337, 2017.

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